

Executive Summary

Overall Verdict: FAIL

Dataset	qc_test_issues.csv
Pass Rate	81.0%
Total Issues	30
Run Date	2026-04-04T12:00:00
Active Checks	11 of 11

Severity Breakdown

Critical		4
Warning		25
Info		1

Issues by Check Category

Check	Critical	Warning	Info	Total
Cross-Column Consistency	2	2	-	4
Spike Detection (Gradient)	-	1	-	1
Coordinate Jumps	-	1	-	1
Range Violations	1	-	-	1
Missing Data	-	-	1	1
Statistical Outliers (Z-Score)	-	1	-	1
Spike Detection (MAD)	-	1	-	1

Issues by Column

dob/depth	2 issues
dob	2 issues
easting/northing	1 issue
doc/dob	1 issue
top/depth	1 issue
depth	1 issue
doc	1 issue
easting	1 issue

Methodology

This report was generated by the DataFlow automated QC engine. The dataset was validated against engineering-grade checks

including:

Basic QC: Range validation, missing data detection, duplicate row/KP identification, KP gap analysis, KP monotonicity enforcement, and statistical outlier detection (z-score and IQR methods).

Engineering-Grade QC: Cross-column physical consistency checks (e.g. DOB vs water depth, DOC vs DOB), spike and gradient detection with KP-aware rate-of-change thresholds, and coordinate sanity validation including bounds checking and consecutive point jump detection.

Each flagged issue includes severity level, affected row and column, and a human-readable explanation of the problem found. Issues are classified as Critical (data integrity broken), Warning (suspicious but usable), or Info (informational).

Detailed Issues (10 of 10)

Row	Column	Severity	Description
6	dob/depth	critical	DOB (48.5m) exceeds water depth (45.1m) -- physically inconsistent
6	dob	warning	dob changes by 47.500 over 0.0500 KP (gradient 950.0/KP) -- exceeds expected rate of 10.0/...
14	dob/depth	critical	DOB (55.0m) exceeds water depth (44.8m) -- physically inconsistent
17	easting/nort	warning	Coordinate jump of 100005m between consecutive rows
5	dob	critical	At KP 0.200: DOB value 48.5m exceeds maximum threshold of 5.0m
8	doc/dob	warning	DOC (1.6m) exceeds DOB (1.4m) -- cover cannot exceed burial depth
15	top/depth	warning	TOP (60.0m) exceeds water depth (45.0m) -- pipe above seabed level
9	depth	info	At KP 0.400: DEPTH value is missing or empty
7	doc	warning	At KP 0.300: DOC value 1.6m is a statistical outlier (z-score: 3.8)
16	easting	warning	easting changes abruptly by 100004.900 between consecutive rows